

Haoyue LIU

Division of Logistics and Transportation, Tsinghua Shenzhen International Graduate School

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EDUCATION

- **Tsinghua University, Shenzhen International Graduate School** Sep.2022 - Present
Ph.D. student, Management Science and Engineering Shenzhen, China
 - Thesis topic: Optimization of Urban Logistics Operations under Uncertainty
 - Advisor: Mingyao Qi (Tsinghua Shenzhen), Sheng Liu (University of Toronto)
- **Beijing University of Posts and Telecommunications, School of Modern Post** Sep.2018 - Jun.2022
Bachelor of Engineering, Logistics Engineering Beijing, China
 - Bachelor of Engineering, grade point ranking is 1/78
 - Honours: Outstanding Undergraduate Student in Beijing, National Scholarship

RESEARCH INTERESTS

- Methodology: Optimization with uncertainty, deterministic optimization, approximation.
- Topics: Supply chain and logistics systems, transportation, urban logistics operations.

SELECTED WORKING PAPERS

- **Traveling Salesman Tardiness**
Haoyue Liu, Sheng Liu. Submitted paper.
- **Recommend-to-Match with Random Supply Rejections: Formulation, Approximation, and Analysis**
Haoyue Liu, Sheng Liu, Mingyao Qi. Major revision at Manufacturing & Service Operations Management.
- **Scheduled Service Network Design Under Demand Uncertainty: A Distributionally Robust Optimization Approach Based on Wasserstein Distance**
Zhiqing Ding[†], Haoyue Liu[†], Daniel Zhuoyu Long, Yongsen Zang, Mingyao Qi. Major revision at Transportation Research Part B.

PUBLICATIONS

- [1] **Haoyue Liu**, Sheng Liu, Mingyao Qi, and Zuo-Jun Max Shen. Joint Workforce Scheduling and Pricing in Crowdsourced Delivery under Uncertainty. *IIE Transactions* (2026): 1-20.
- [2] Jiemin Zhao[†], **Haoyue Liu[†]**, Xin Wang, and Mingyao Qi. A Mobile Clinic Routing and Scheduling Problem in Vaccination Outreach Service. *Omega* (2026): 103534.
- [3] Di Wang, **Haoyue Liu^{*}**, and Yuming Li. Intelligent Weight Generation Algorithm Based on Binary Isolation Tree. *Engineering Applications of Artificial Intelligence* 109 (2022): 104604. (The first author except supervisor)

OTHER EXPERIENCE

- First Prize (Top 1) of the 29th Tsinghua "Hua Luogeng Cup" Mathematical Modeling Competition in May 2023
- Excellence Award (Top 5) of SF-X Smart Supply Chain Campus Elite Challenge ([sf-express](#)) in Aug. 2023
- Second Prize (Top 10%) of "Huawei Cup" 20th China Postgraduate Mathematical Modeling Competition in Nov. 2023
- Teaching Assistant, Distribution System Modeling and Analysis (2023 Fall)
- Conference oral presentation: POMSHK (Jan. 2025, Jan. 2026); Informs conference on service science (Jun. 2024)

ADDITIONAL INFORMATION

- **Languages:** Mandarin (native), English (conversational)
- **Technical Skills:** Python, Java, MS Office
- **Interests:** Singing and reading (especially history and philosophy)